

Erdős Problem #1049

EP-1049

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $t > 1$ be a rational number. Is $\sum_{n=1}^{\infty} \frac{1}{t^{n-1}} = \sum_{n=1}^{\infty} \frac{\tau(n)}{t^n}$ irrational, where $\tau(n)$ counts the divisors of n ?